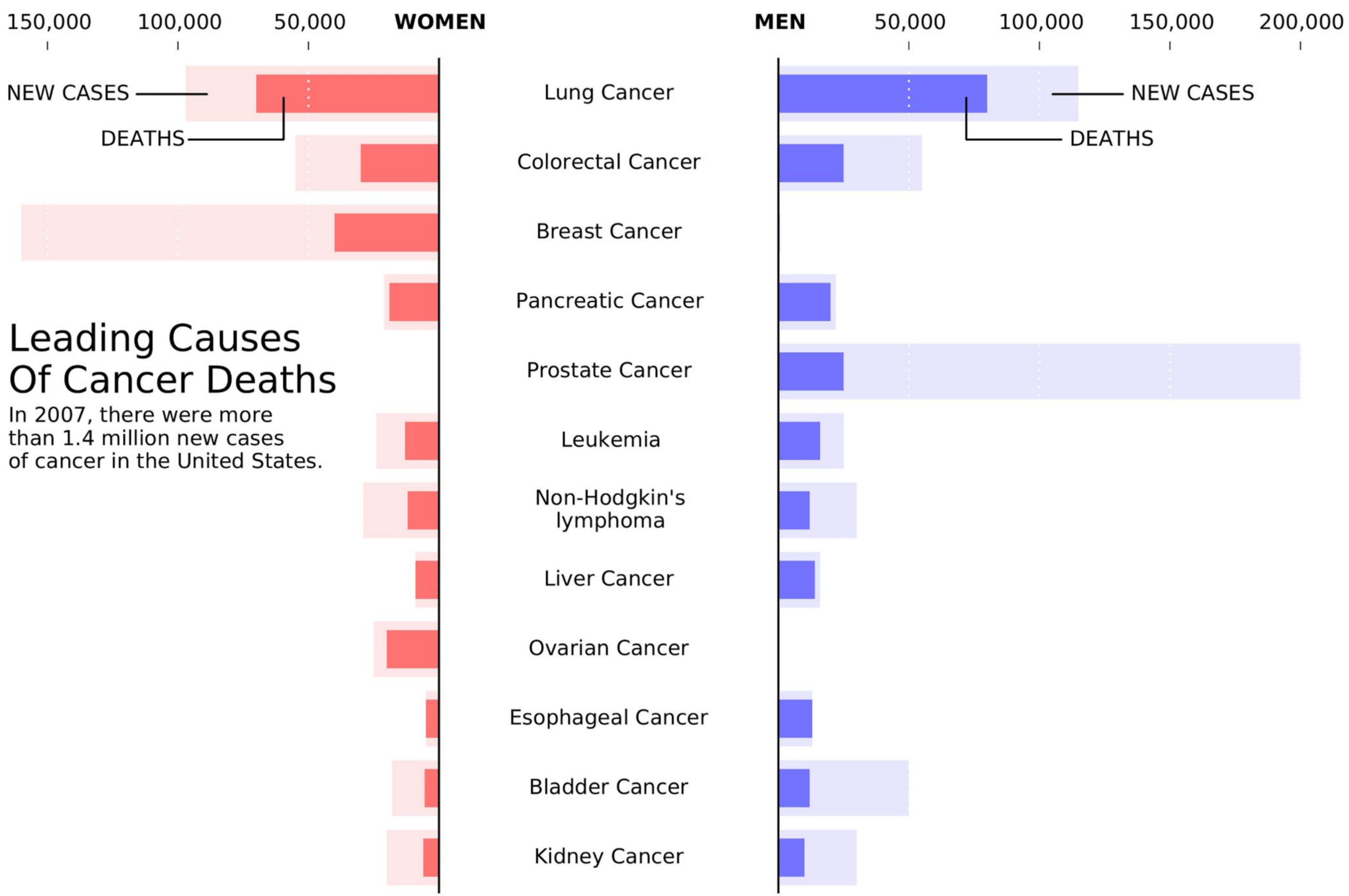
Rules for better figures

Lars Pastewka

Figures are important…

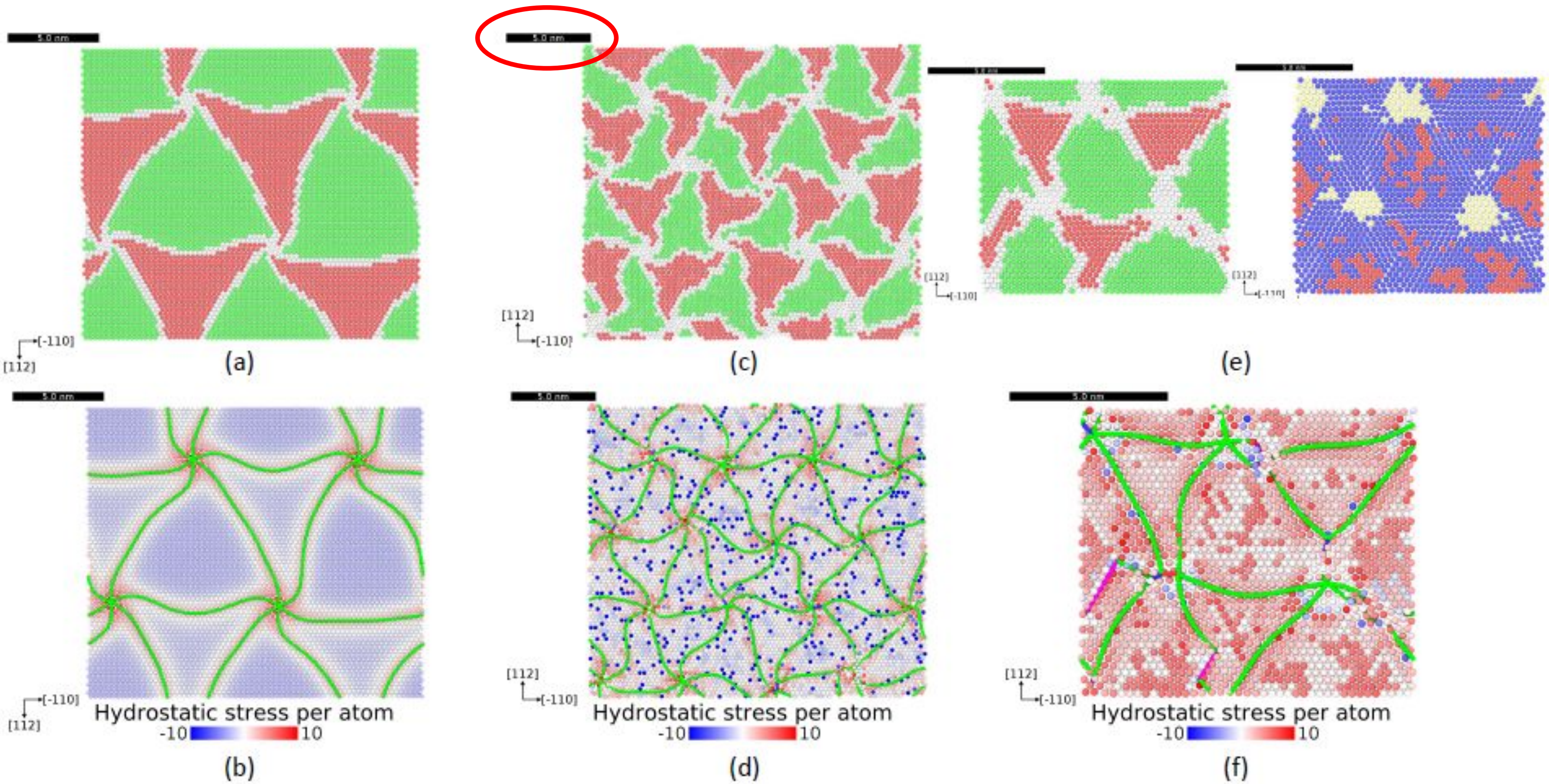
- If someone decides to like the title of your paper:
  - 1% of those will actually read it
  - 5% of those may read the abstract
  - 10% may read figure captions
  - …but 100% will look at figures
- Your story should be in the figures
- From a referee report…

  Fig. 2: The arrow with which "first asperity" points to the line is a bit small. In addition, the authors may want to add more descriptors to the figures so that one does not have to read the captions. (I, personally, hate reading captions.) They could place the following comments above the legends (a) hard-wall repulsion, (b) soft repulsion, (c) adhesive interactions

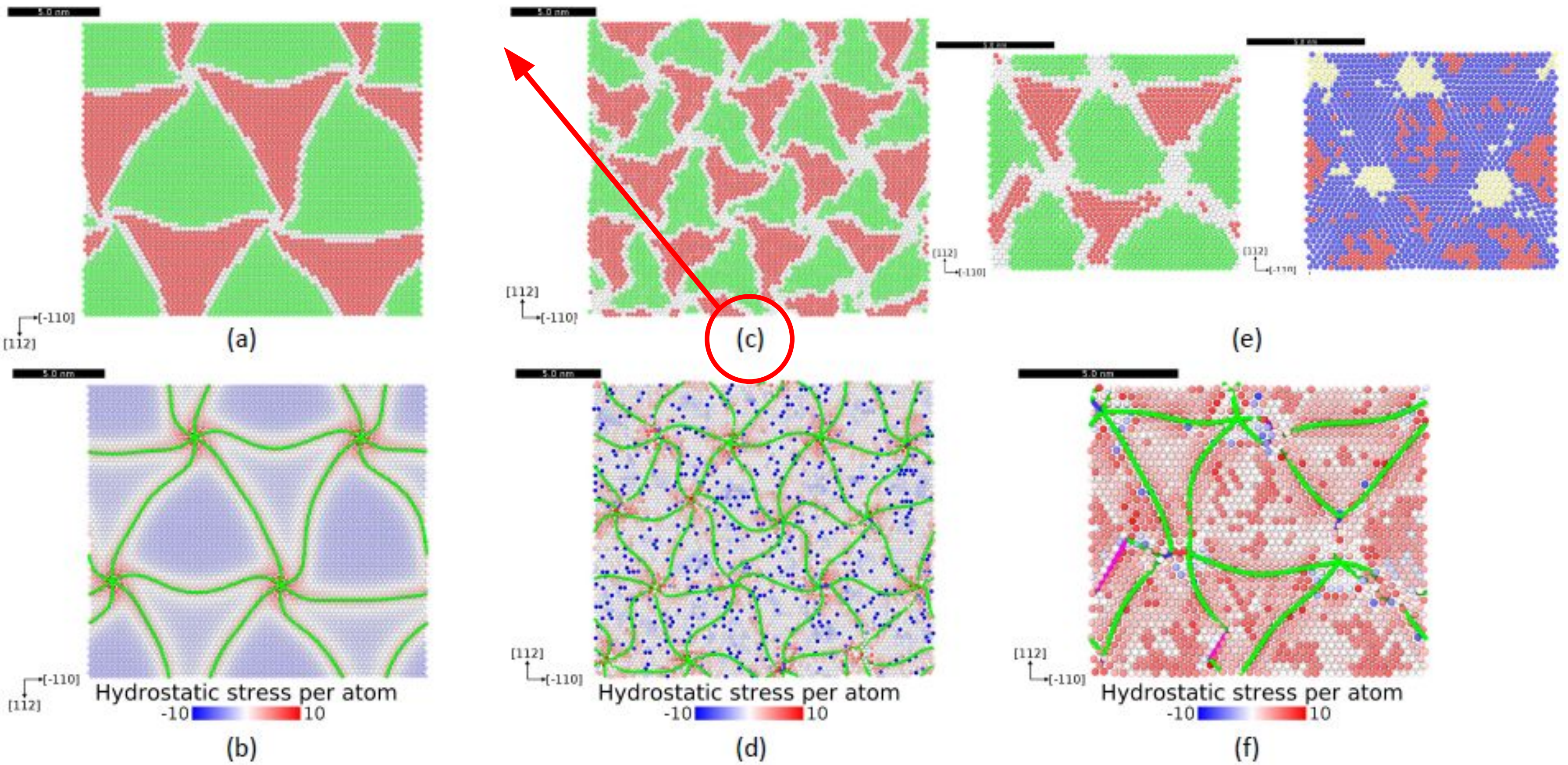


Rougier, Droettboom, Boume, Ten simple rules for better figures, PLOS Comput. Biol. 10, e1003833 (2014)

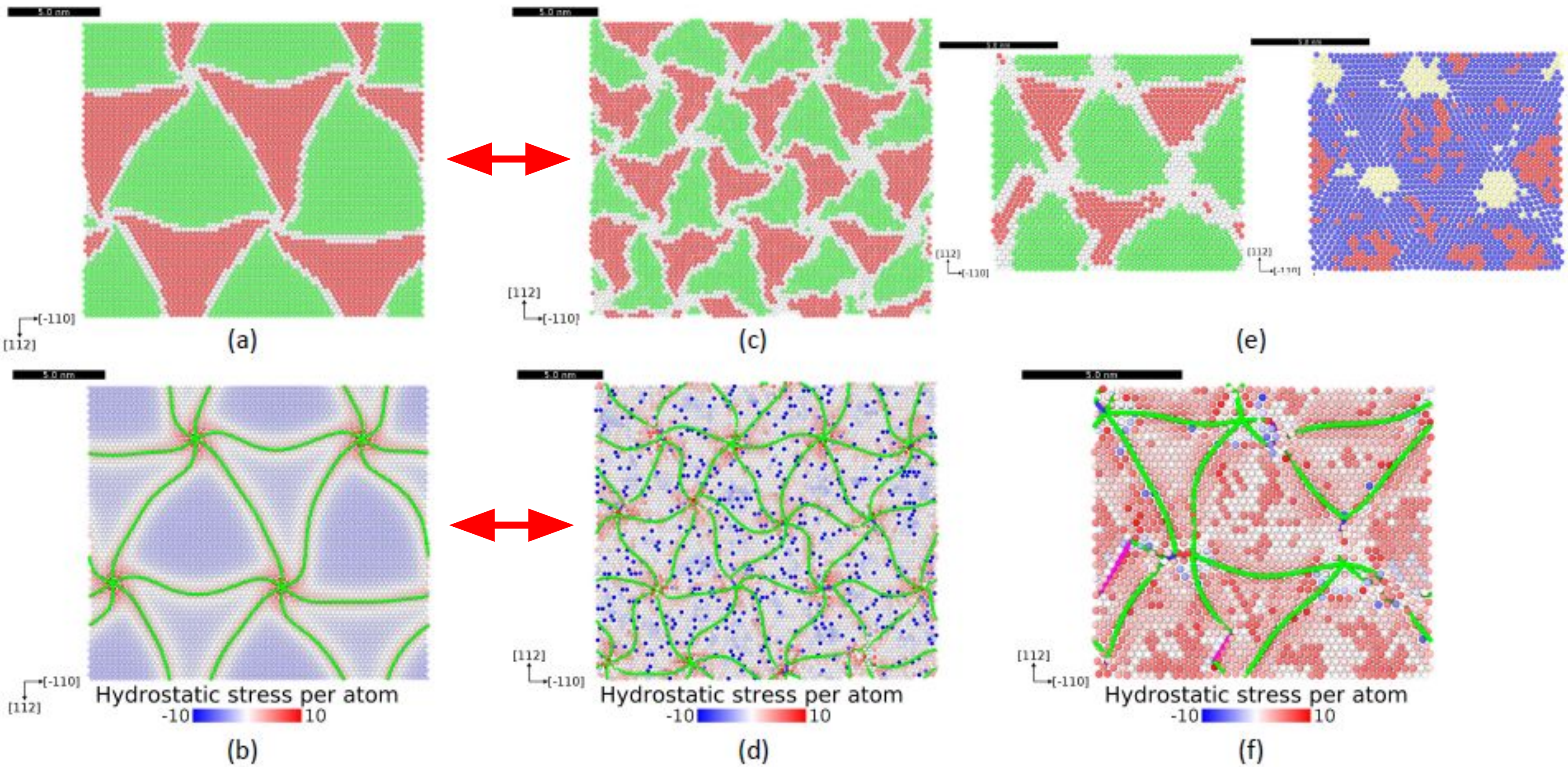
1 You should be able to read the labels



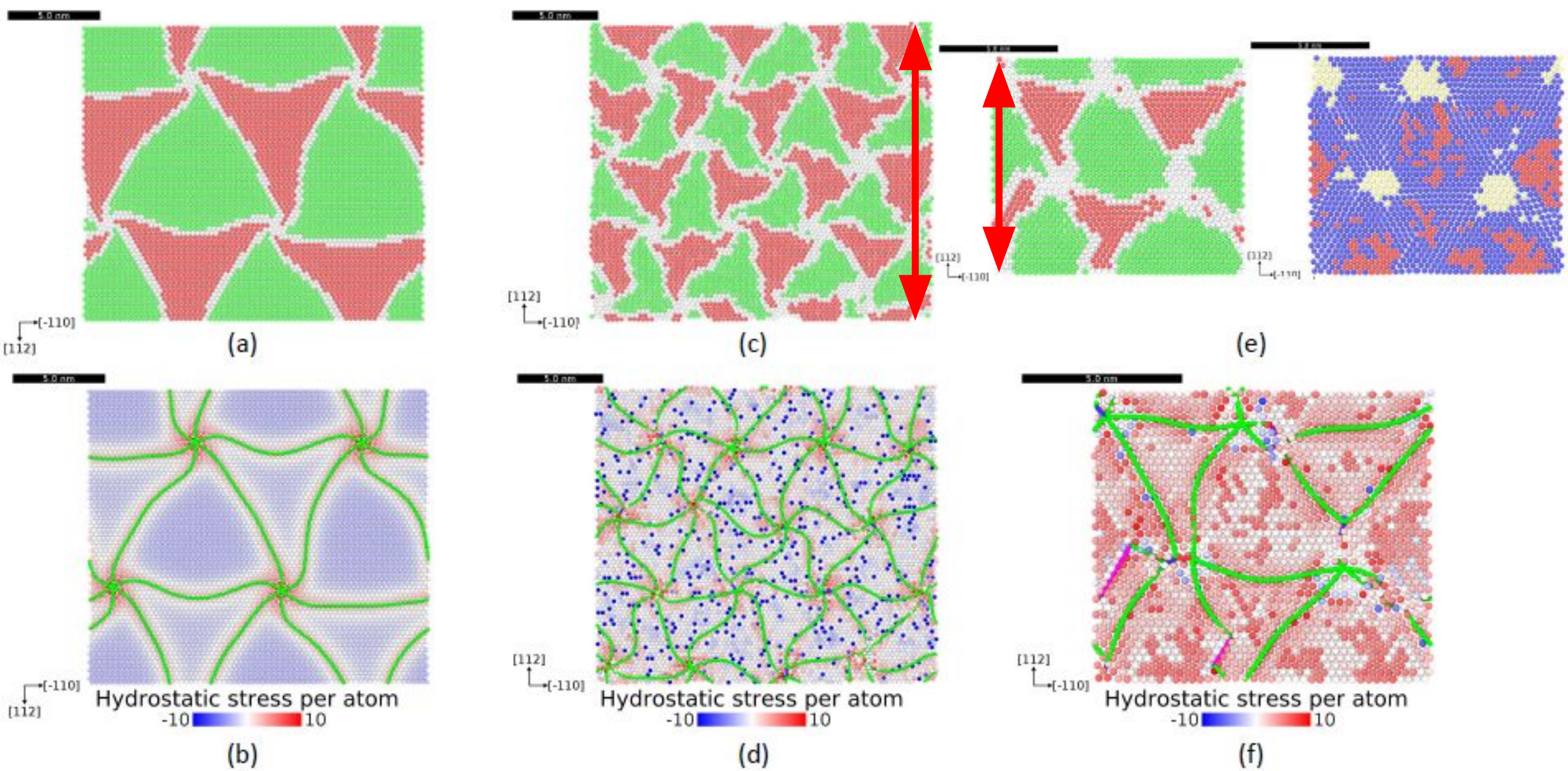
2 Panels should be numbered, top left



3 Use space wisely



4 Use identical scales



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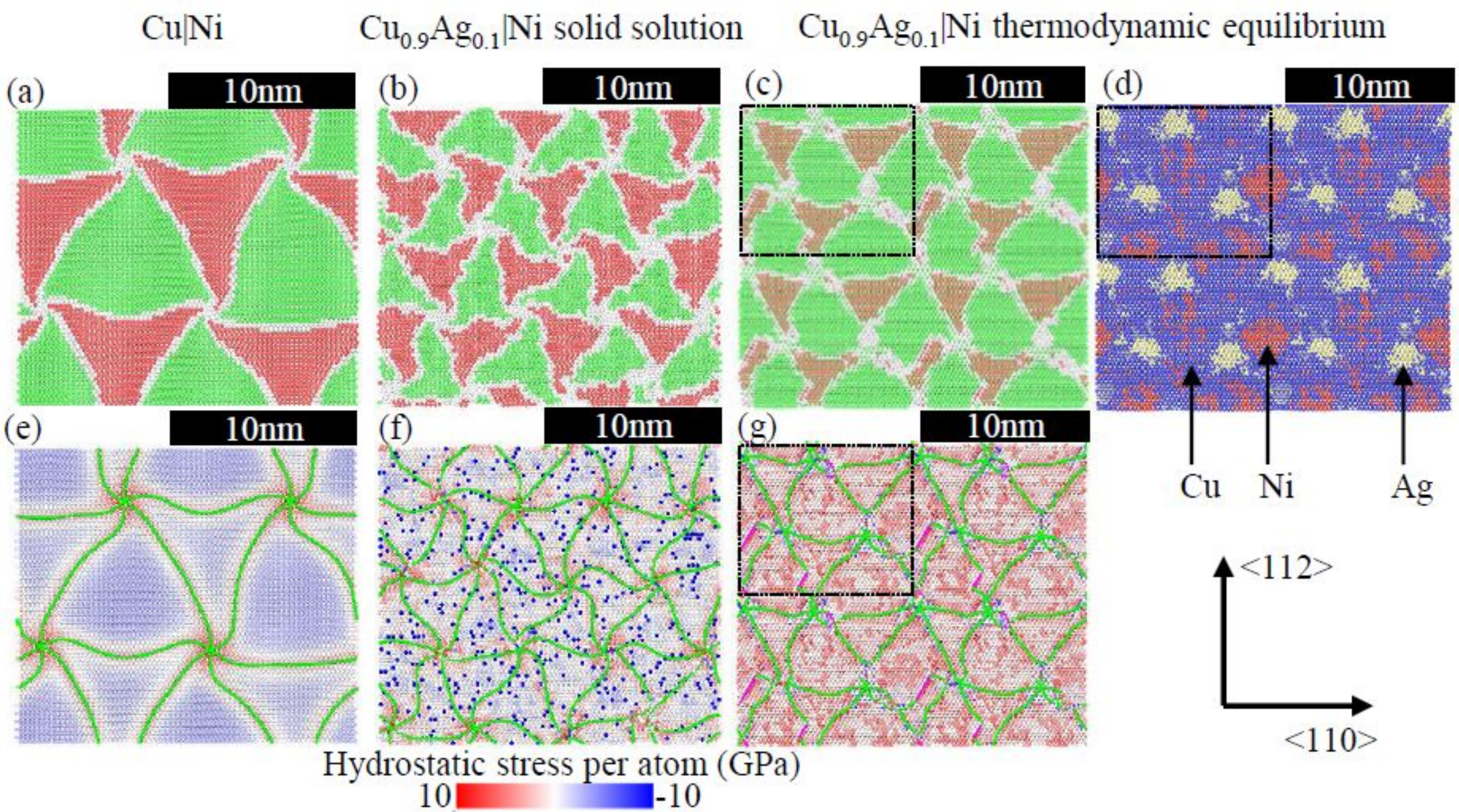

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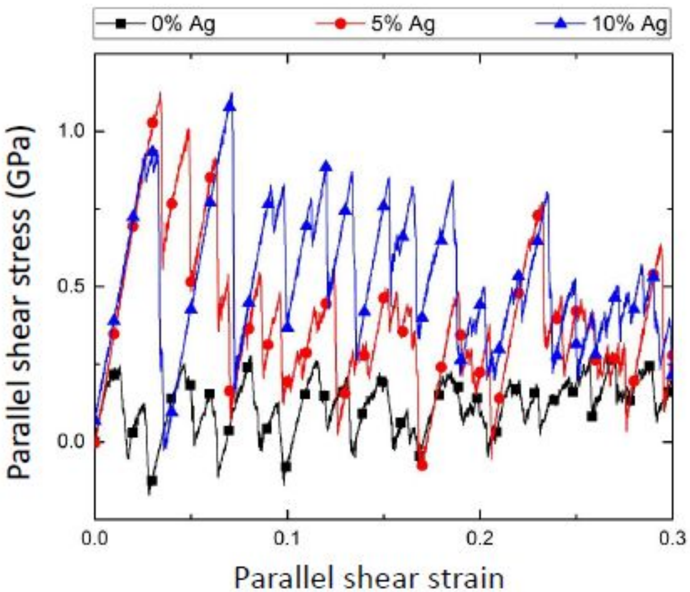

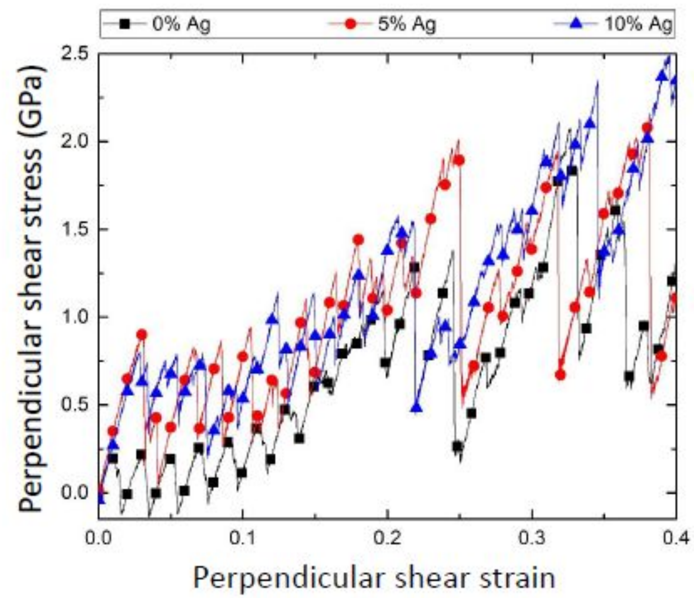

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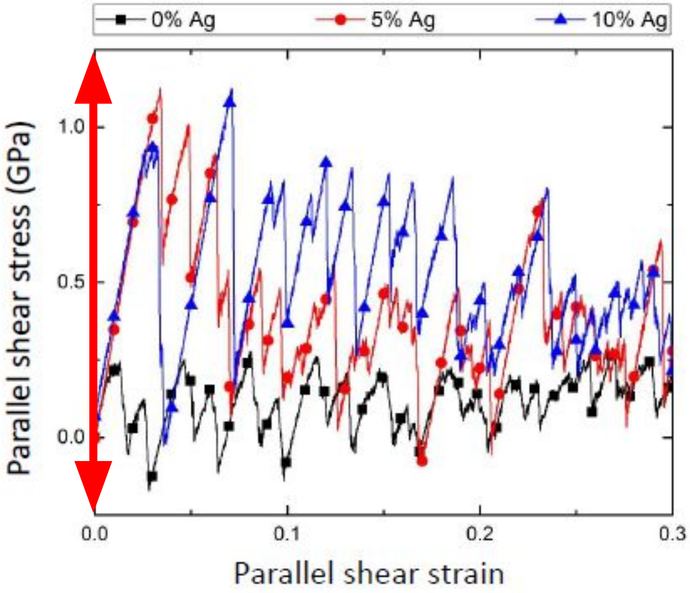
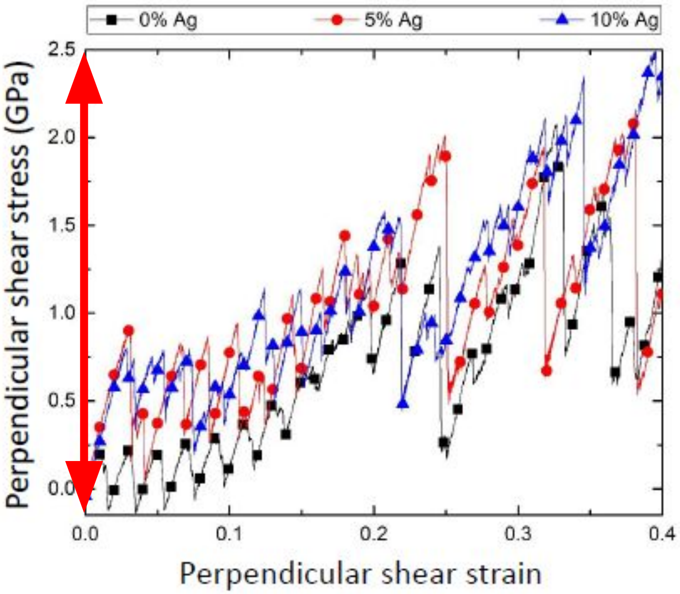
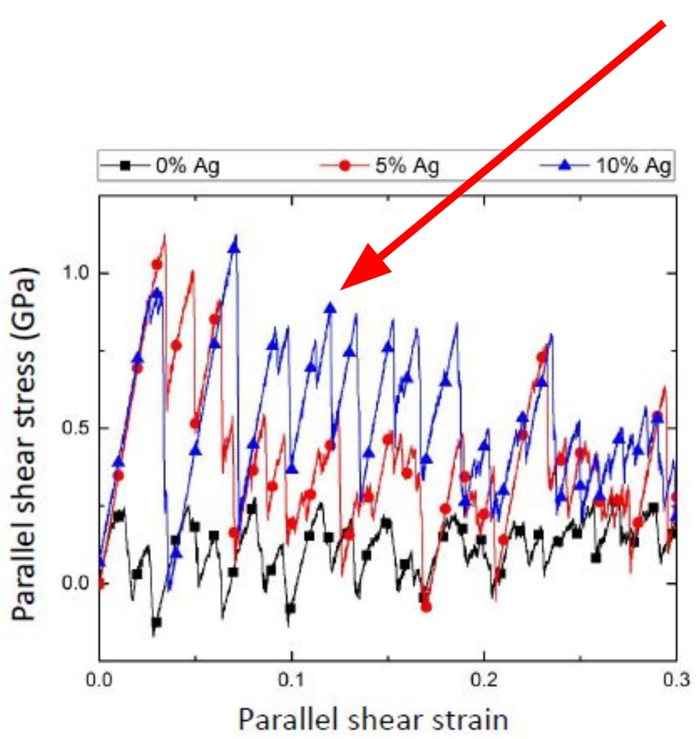
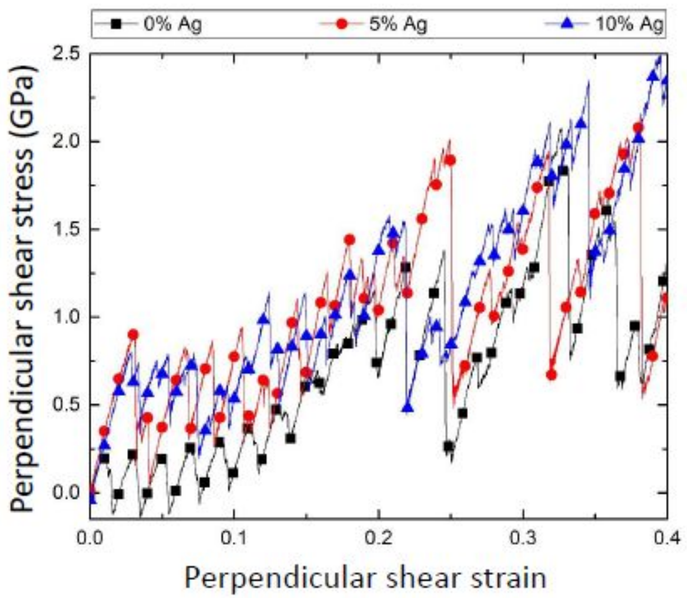




5 Points are data, lines LINEAR interpolation





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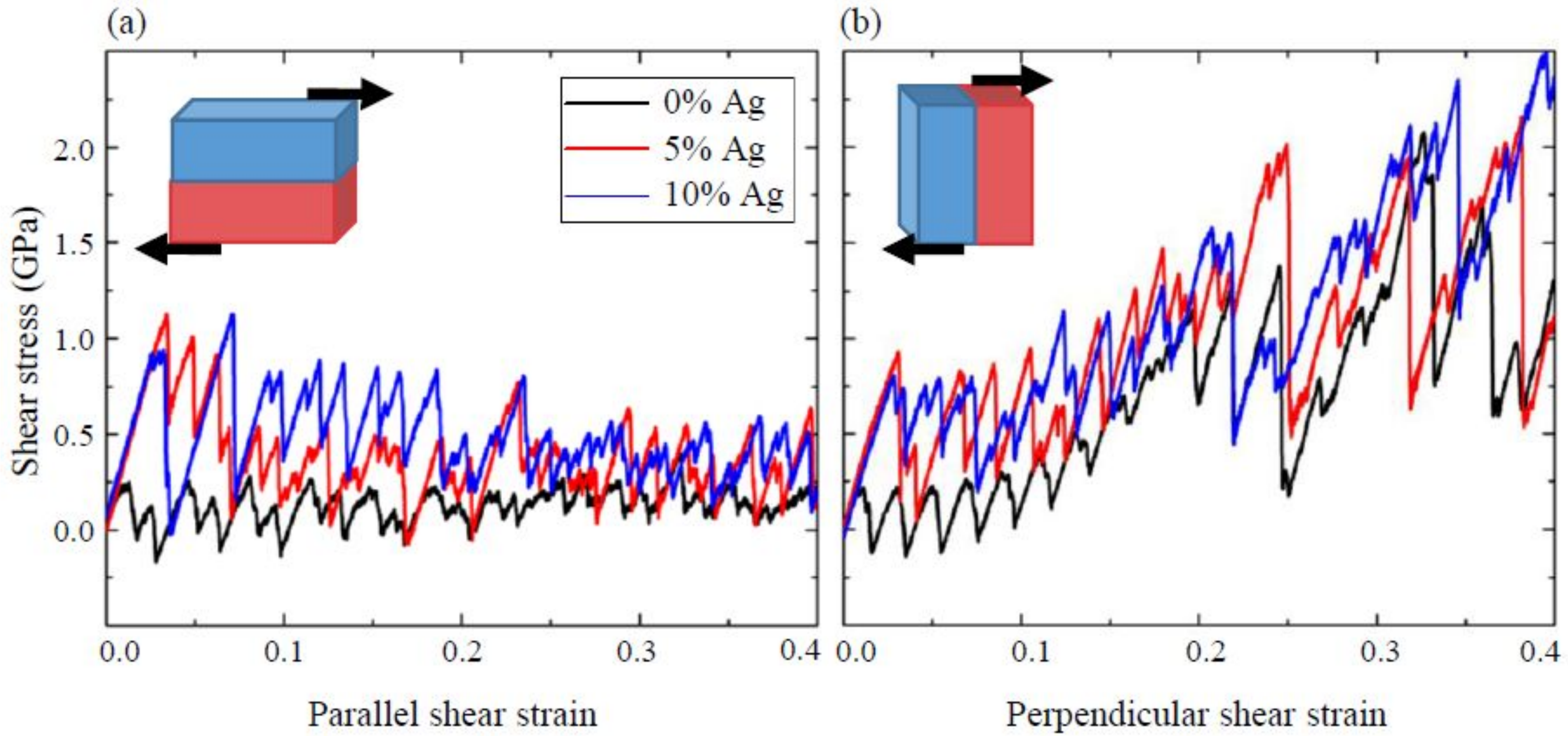

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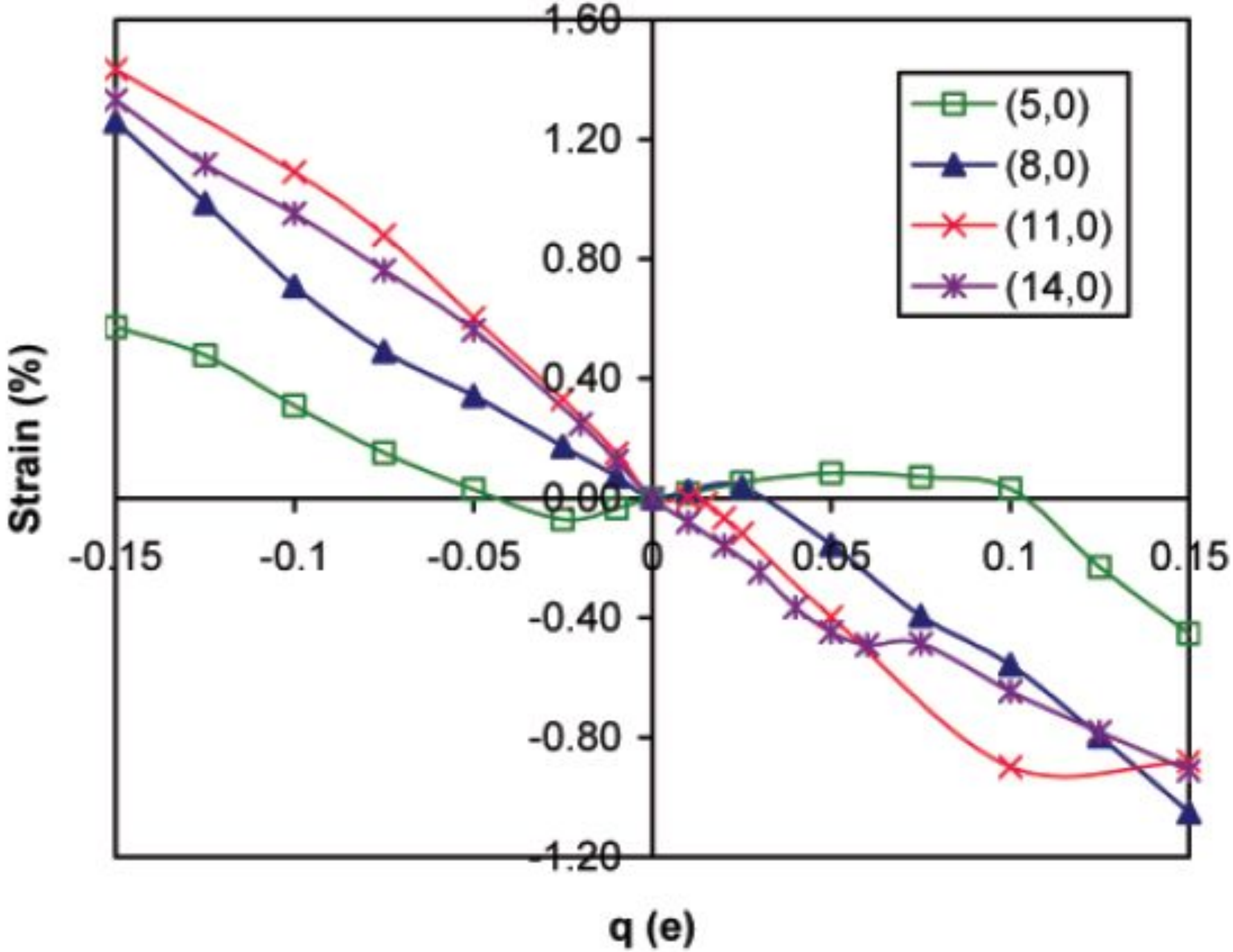
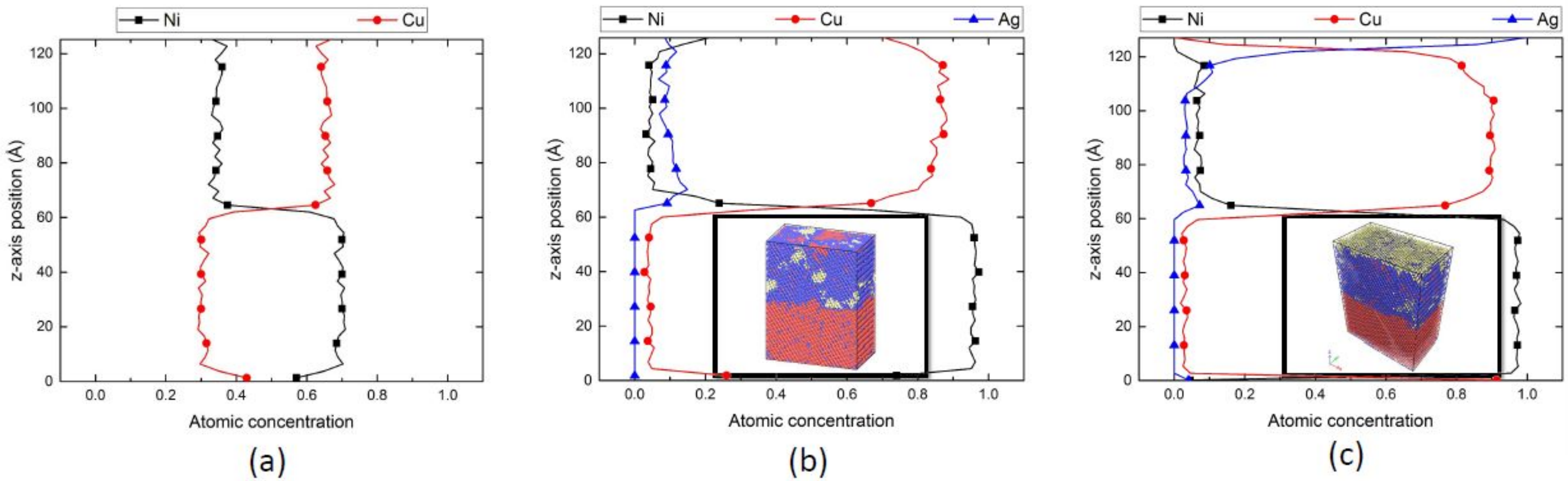
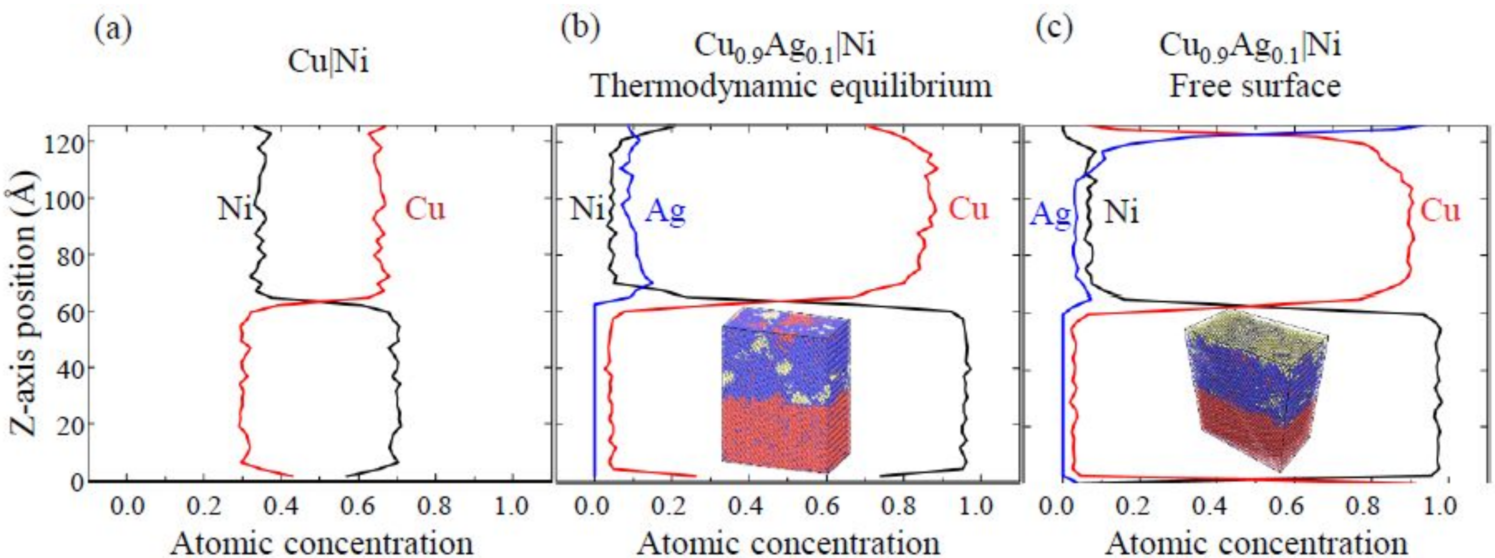


Sun, Kürti, Kertesz, Baughman, J. Am. Chem. Soc. 124, 15076 (2002)

6 Captions trump legends





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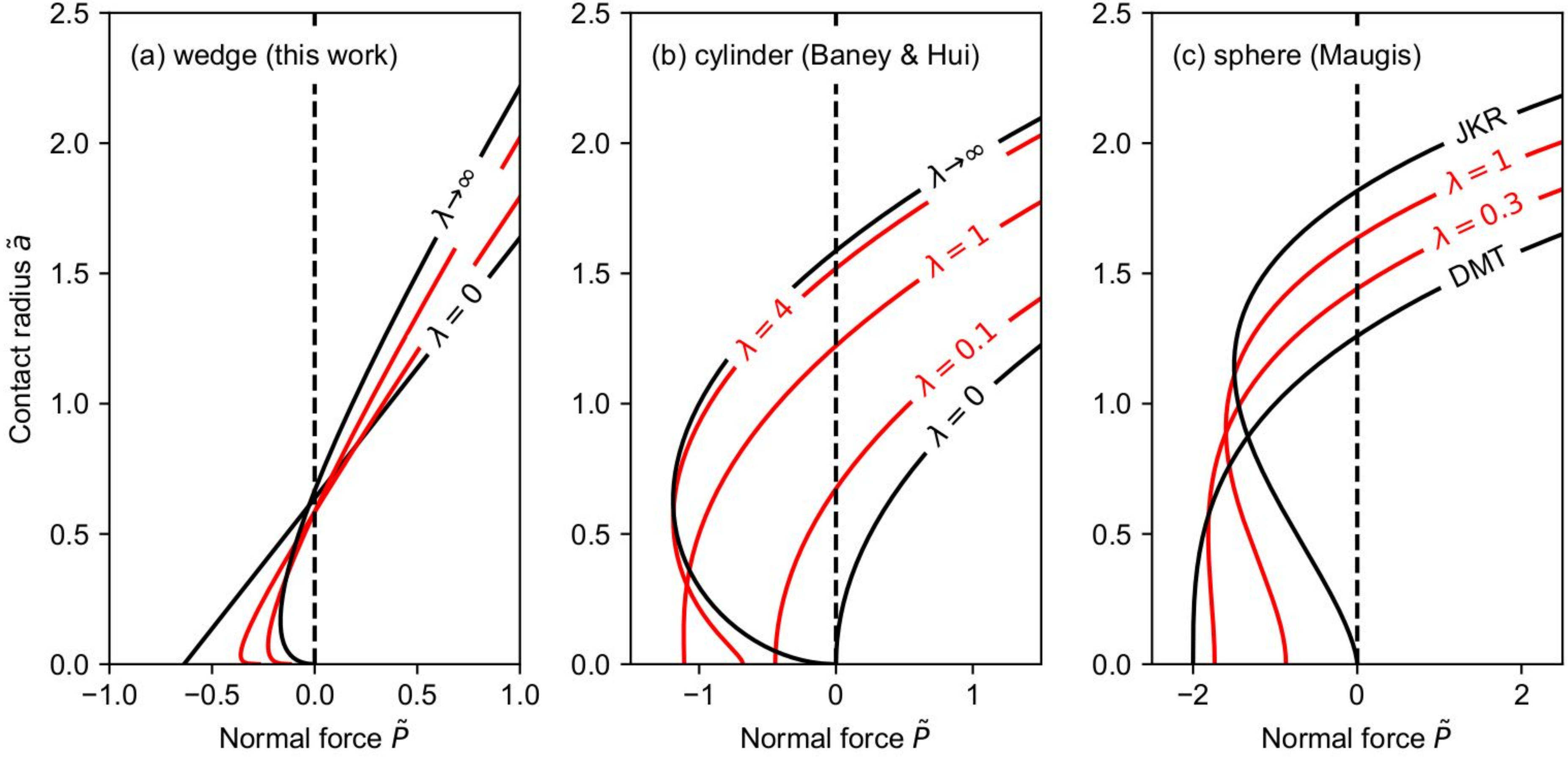
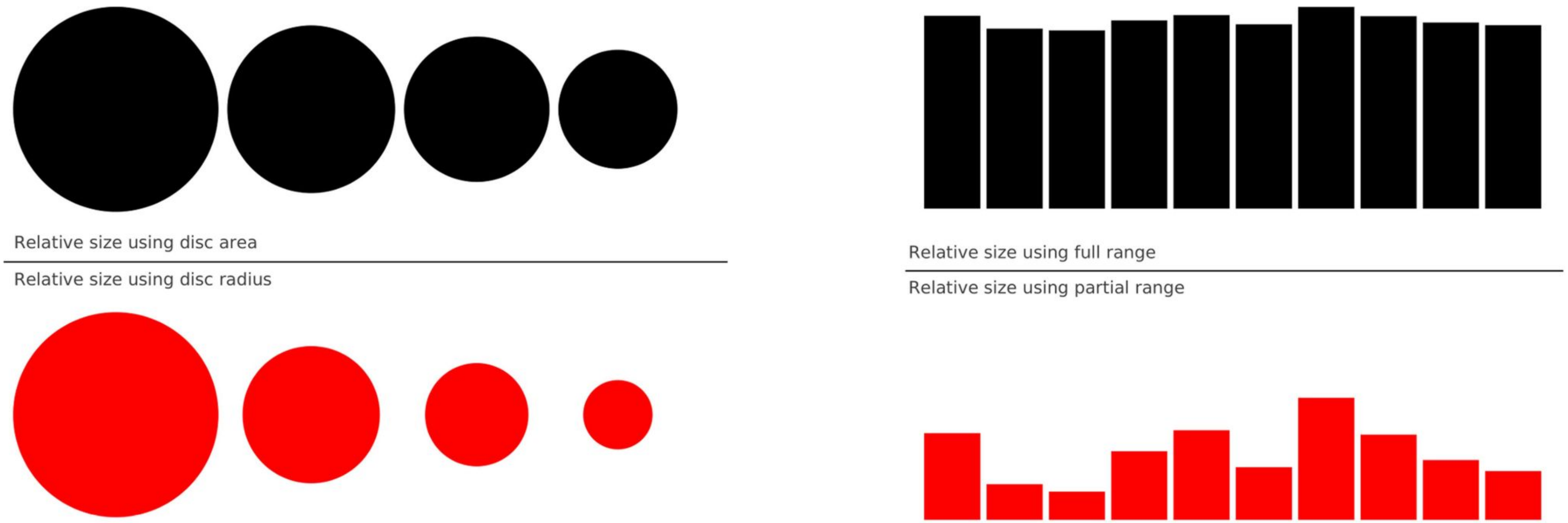


7 Do not mislead the reader



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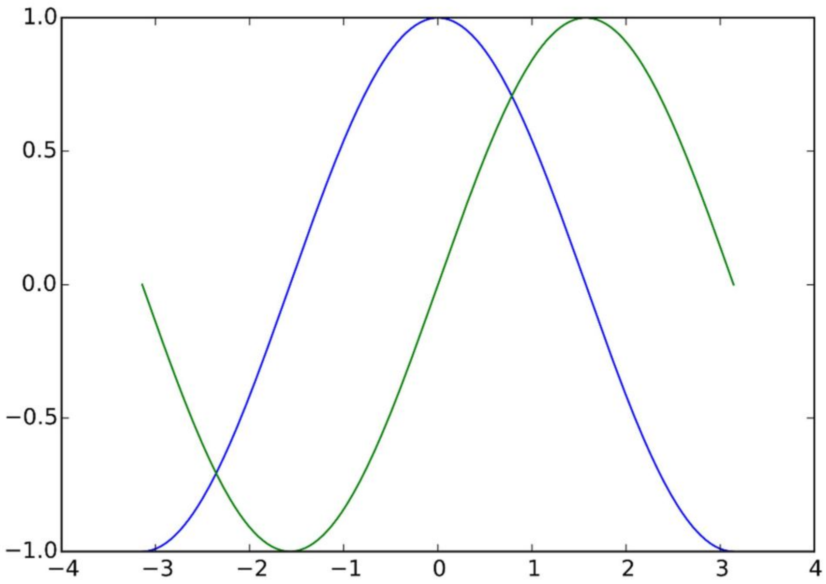
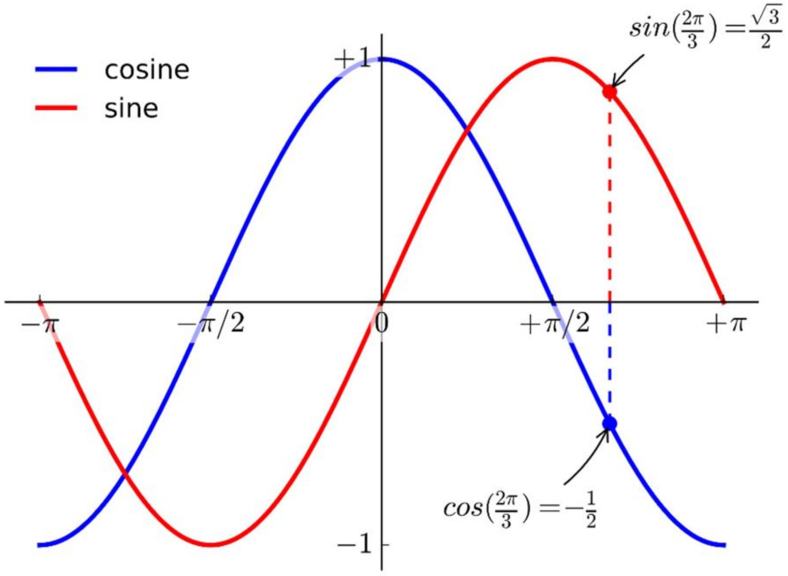
8 Do not trust the defaults





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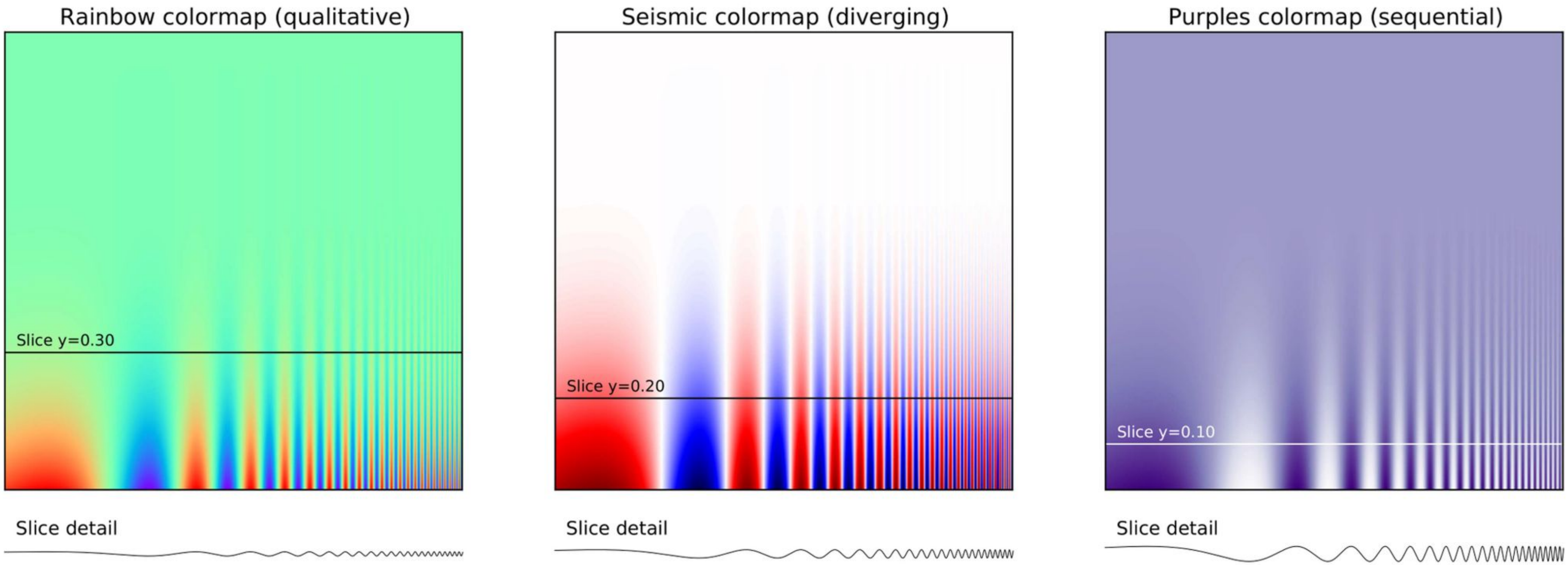
9 Think about color maps



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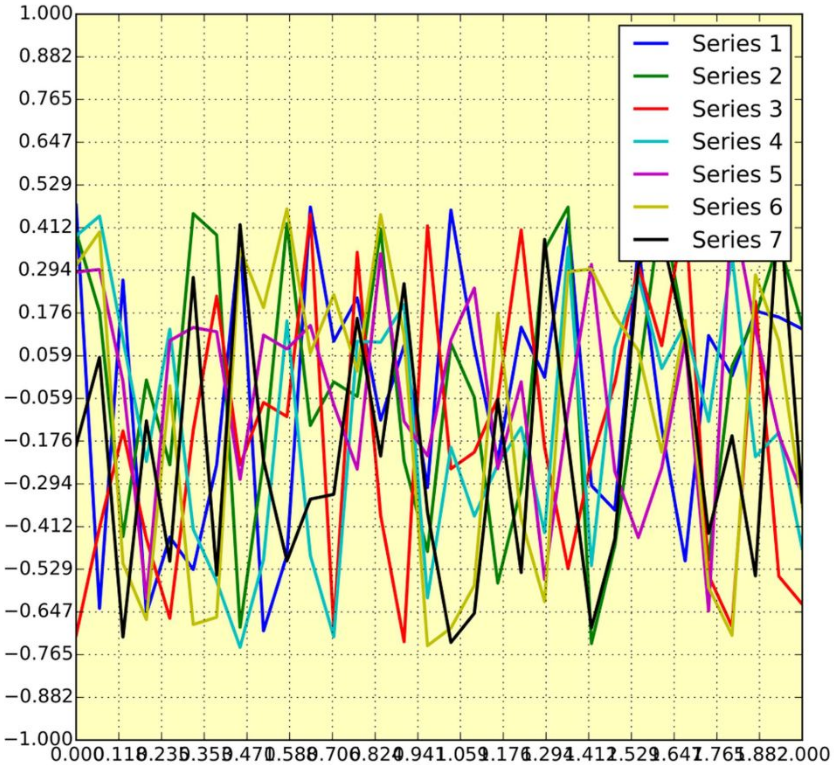
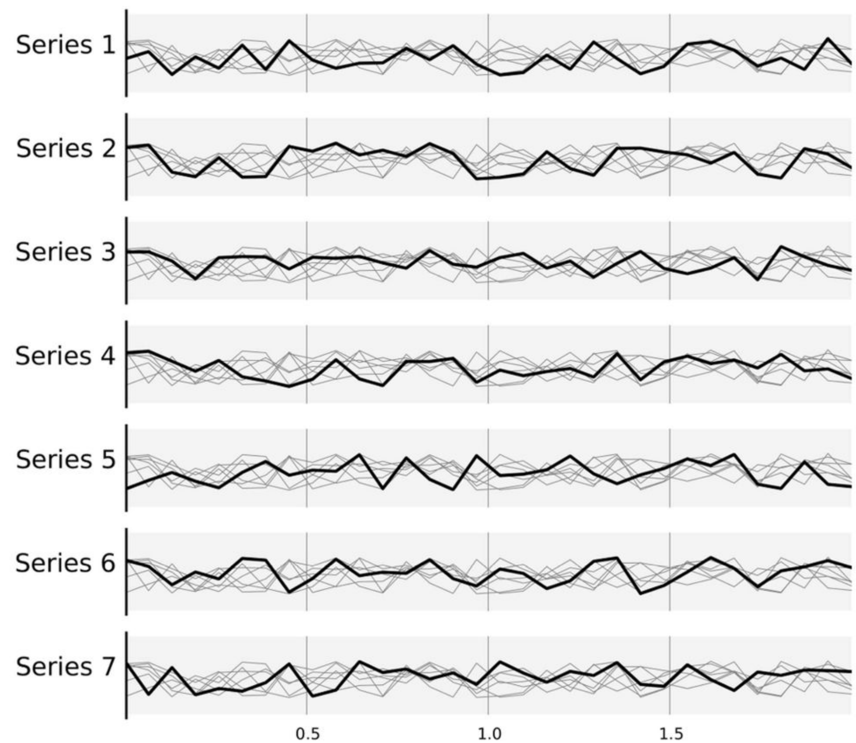
10 Avoid chartjunk





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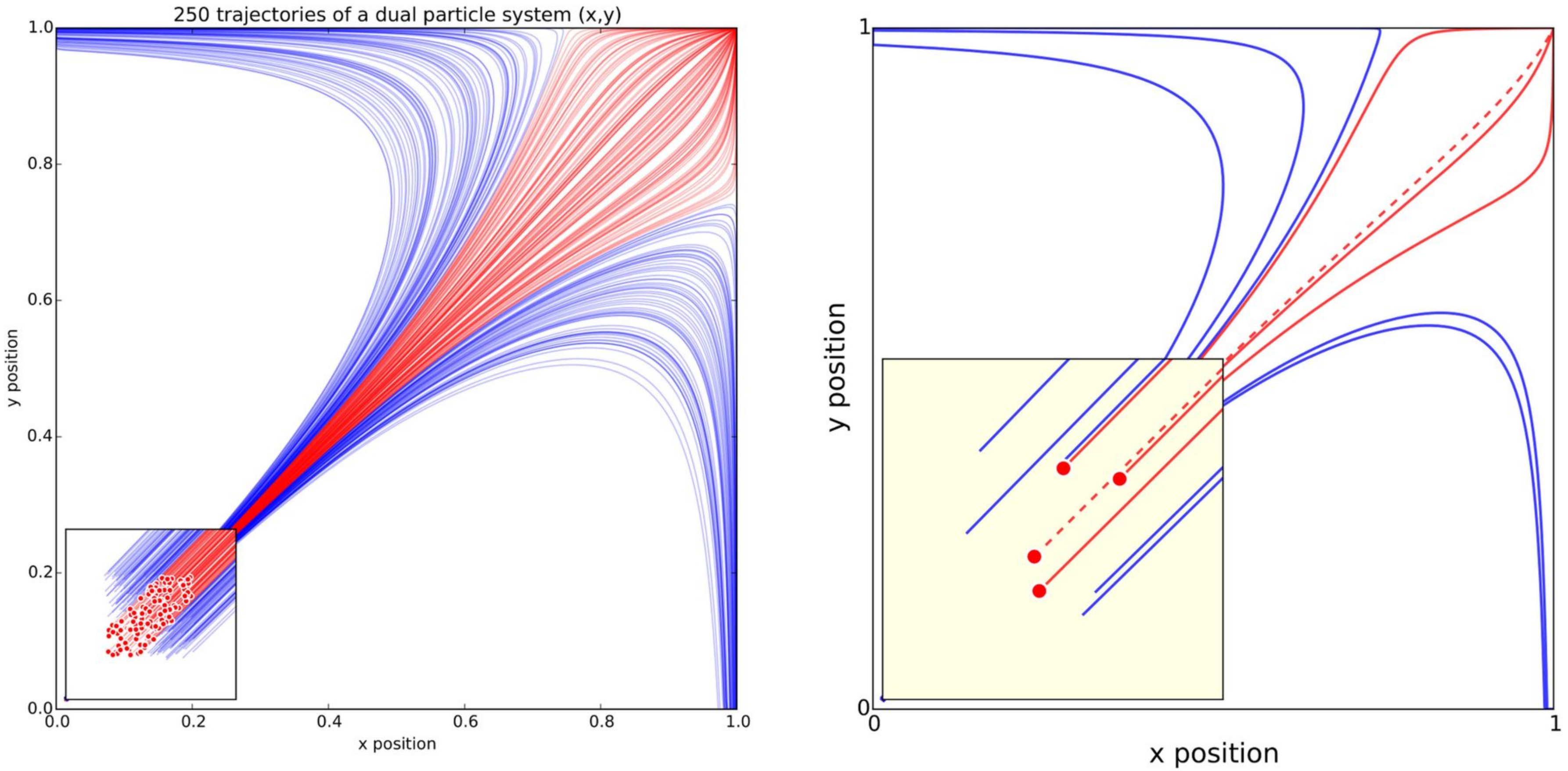
11 Adapt your figure to the audience



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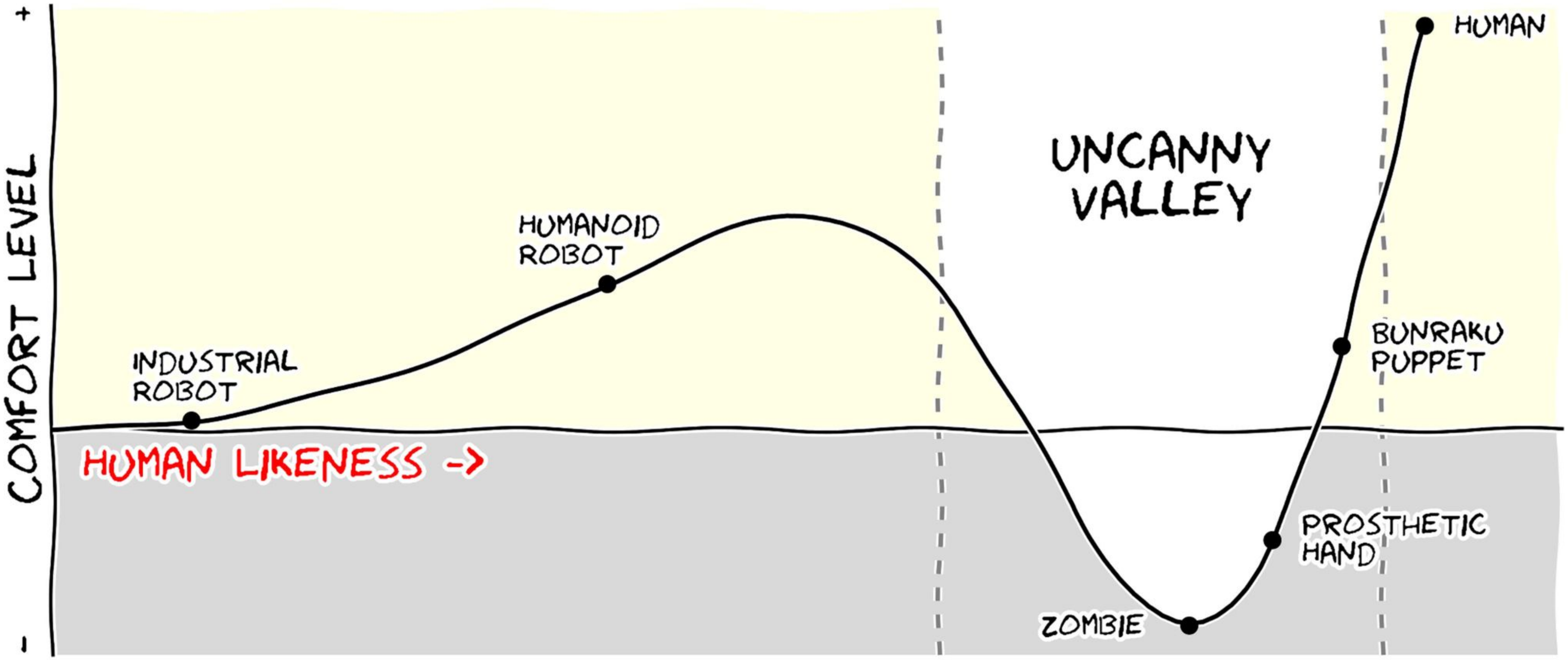
In any case: Message trumps beauty



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Units

Specific units in parenthesis

- *σ* (GPa)
  or better: Shear stress *σ* (GPa)

DIN (acceptable)

- *σ*/GPa
- *σ* in GPa

Absolutely not

- *σ* [GPa]

Use the right (combination of) tools

Free

- Jupyter Lab - http://jupyter.org/
- matplotlib - http://matplotlib.org/
  - Seaborn - https://stanford.edu/~mwaskom/software/seaborn/
- Inkscape - https://inkscape.org/
- Gimp - https://www.gimp.org/
- IPE - http://ipe.otfried.org/
- Blender - https://www.blender.org/

Commercial

- Origin
- PowerPoint
- DataGraph (if you have a Mac): http://www.visualdatatools.com/DataGraph/
- Illustrator

Web

- http://colorbrewer2.org/ a